PARTH MAHAJAN

Robotics Software Engineer, SLAM & Localization

Boston, MA | mahajan.parth@northeastern.edu | +1-617-953-7065 | LinkedIn | Projects | GitHub | Blogs

EDUCATION

Northeastern University, *MSc in Robotics, GPA 4.0/4.0* Boston, MA | **Sep 2024 to Dec 2026 (Expected)**
Courses: Mobile Robotics, Computer Vision, Deep Learning, Control Systems, High Performance Computing

EXPERIENCE

Crabi Robotics, *Localization and Navigation Intern* Boston, MA | **May 2025 to Dec 2025**

- Built a multi-session V-SLAM pipeline (C++, ROS 2, GTSAM) for low-texture ship hulls, fusing loosely coupled wheel and IMU odometry, AprilTag loop closures and opportunistic visual constraints at <15 cm trajectory RMSE.
- Prototyped a speckle-laser-based optical odometry system that estimates motion from optical flow fields of spectral interference patterns, reaching about 1% RMSE trajectory error on closed-loop trajectories.
- Replaced skid-steer with 4-wheel independent-steering forward and inverse kinematics, cutting tread wear 30%.
- Created a physics-based Isaac Sim pipeline for sensor-faithful data, mapping validation and client demos.

IIIT Hyderabad, Robotics Research Center, *Research Assistant* Hyderabad, India | **May 2023 to Aug 2024**

- Extended LeGO-LOAM by adding GNSS measurements as position factors to its factor-graph backend, jointly optimizing LiDAR odometry and global position to reduce long-term drift.
- Published Guided Polynomial Diffusion, which pairs DDIM sampling with Bernstein polynomials for safe trajectory generation in manipulators and cuts diffusion time 10× against EDMP. [Paper]
- Developed and tuned Pure Pursuit and Stanley controllers in C++ on a 1-ton Mahindra electric vehicle, holding mean cross-track error to 0.15 m. [Video]

Addverb Limited, *Robotics Software Engineer L2* Noida, India | **Aug 2021 to Aug 2023**

- Wrote an eye-in-hand and eye-to-hand calibration pipeline in C++ and OpenCV, estimating camera intrinsics and robot-camera extrinsics with <0.3 px test-set reprojection RMSE.
- Developed low-level C++ ROS drivers for the LiDAR and ultrasonic sensors that feed the AMR safety modules, and migrated core AMR200 perception and navigation components from ROS 1 to ROS 2.
- Accelerated SegNet vision models with TensorRT, OpenVINO and DV SDKs through post-training quantization to UInt8 and FP16, raising throughput 2 to 5× while maintaining precision.
- Led an end-to-end UR5 bin-picking system with deep-learning 6-DoF grasp pose estimation across 1.5K+ SKUs.

PROJECTS

Certifiable Measurement Selection for Calibration | *Convex Optimization, OpenCV, Kalibr, Isaac Sim* GitHub

- Built an uncertainty-aware calibration pipeline with Fisher information and Schur-complement marginalization, cutting held-out reprojection RMSE 15%, from 0.883 to 0.752 px, versus random views on real ChArUco data.
- Implemented E-, A- and D-optimal view selection using greedy and Frank-Wolfe optimization with duality-gap certificates, achieving solutions within 2.5% of exhaustive-search E-optima on validated benchmark instances.

Wheel Odometry and IMU Sensor Fusion | *C++, Eigen, ROS 2, EKF* GitHub

- Implemented an 8-state EKF in C++, Eigen and ROS 2 with analytic Jacobians and a constant-acceleration model; on the outdoor Linkou dataset (247 m), it cut ATE against GPS from 19.58 m (wheel odometry) to 1.72 m.

LM-DROID-SLAM: Levenberg-Marquardt Optimization | *Python, PyTorch, CUDA, Visual SLAM* GitHub

- Profiled DROID-SLAM using cProfile on TartanAir to identify computational bottlenecks, attributing 35% of runtime to the update operator and 30.7% to graph aggregation, versus just 7.5% to dense bundle adjustment.
- Integrated a differentiable Levenberg-Marquardt solver with adaptive damping into DROID-SLAM over pose and inverse depth; TartanAir benchmarks showed that accuracy degrades without end-to-end retraining.

JEV Smart Robot Control: Localization Recovery | *ROS 2, Nav2, AMCL, Gazebo* GitHub

- Integrated the state-of-the-art JEV model into a ROS 2 Nav2 stack as a smart decision layer to recover from AMCL localization failures, localizing correctly in 97% of trials across 120+ random spawn points.

TECHNICAL SKILLS

Languages and Tools: C++, Python, ROS 2, Eigen, OpenCV, Open3D, Docker, Git, Linux, CUDA
Robotics and Sensors: GTSAM factor graphs, Particle Filters, EKF, Sensor Synchronization, Optimization (LM, GN), Intrinsic and Extrinsic Calibration, VO, VIO-SLAM, Isaac Sim, Gazebo, RViz2, Camera, Ultrasonic, IMU, GNSS
AI/DL: PyTorch, NumPy, scikit-learn, SciPy, Matplotlib, W&B, TensorRT, Segmentation, Deep Optical Flow